

Testing for non-stationarity in multivariate extremal clusters with application to Spanish droughts

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1 | INTRODUCTION

From confirmation report We now consider settings in which the extremal dependence structure is non-stationary, and may change over time. For example, meteorological variables observed across spatial locations may exhibit changes in extremal dependence due to shifts in climate regimes, while biomarker measurements collected longitudinally for individuals may display structural changes associated with disease progression, treatment effects, or lifestyle changes (Katz 2010; Nicolet, Eckert, Morin, & Blanchet 2018; Panagoulas, Tsihrintzis, & Virvou 2025; Talento, Richards, Pinto-Orellana, Huser, & Ombao 2025). In such settings, it is of interest to detect changepoints or structural breaks in the extremal dependence structure.

In the existing framework, the multivariate “groups” could, in principle, represent different time periods. However, this method was not developed to accommodate temporally evolving dependence structures within an already multivariate setting (e.g. spatial), and we assumed that the data within each group was stationary. This is limiting in applications where tail dependence is expected to vary over time. In environmental contexts, for example, climate change may strengthen or weaken extremal dependence across an entire spatial domain, as suggested by [Zhong, Huser, and Opitz \(2022\)](#), who found that recent Southern European heatwaves have been more widespread than in the past. More generally, several types of structural change may occur. In some situations the extremal dependence structure may shift globally across all sites, corresponding to a regime change affecting the entire system. In other settings, only a subset of sites may change their extremal dependence behaviour, for example by switching between clusters of locations with similar tail dependence characteristics. Discarding the stationarity assumption and allowing for such temporal changes in extremal dependence is therefore a natural and important extension of our current work. It would enable us to identify time periods with distinct spatial patterns of tail dependence, as well as the specific changepoints at

⁰**Abbreviations:** ANA, anti-nuclear antibodies; APC, antigen-presenting cells; IRF, interferon regulatory factor

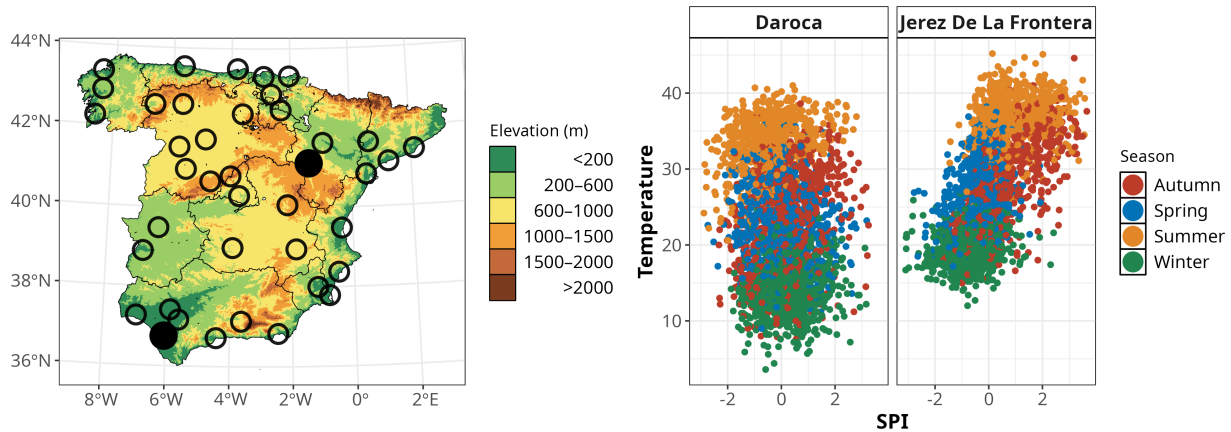


FIGURE 1 Left: elevation profile of mainland Spain and locations of the 40 meteorological stations, with Daroca and Jerez de la Frontera highlighted. Right: weekly maximum temperature against the 13-week drought index $D = -\text{SPI}$ at the two highlighted stations, with observations coloured by season.

which these transitions occur. Therefore, a key focus of this work will be the development of methods for changepoint detection in multivariate extremal dependence structures, building on the divergence-based framework introduced in our first paper.

There has been some previous work on changepoint detection in multivariate dependence structures, whether in the context of extremes or more generally. For example, [McGonigle, Pawley, Richards, and Rohrbeck \(2026\)](#) developed a non-parametric sliding window approach for detecting changepoints in pairwise extremal dependence. Also, [Alouini and Davison \(2026\)](#) developed a Bayesian clustering algorithm for multivariate extremes which utilises a mixture of Dirichlet distributions, and they show how it can be used to temporally cluster negative returns for several S&P 500 stock sectors, thereby identifying periods of time with similar extremal dependence structures across the sectors. In [Agarwal, Eckley, and Fearnhead \(2025\)](#), a likelihood-based method for detecting temporal changepoints in spatio-temporal data is developed, and applied to daily wind speed data in Ireland, similar to the data we used in our first paper. We hope to build on these existing approaches and also the divergence-based clustering framework developed in our paper to develop a method for detecting changepoints in multivariate extremal dependence structures, which can be applied to a wide range of datasets and applications.

2 | SPANISH TEMPERATURE AND DROUGHT DATA

Desertification is one of Spain's most pressing environmental challenges. [Martínez-Valderrama et al. \(2016\)](#) found that 20% of Spanish territory was degraded and an additional 1% was undergoing an active process of environmental deterioration during 2000–2010. While part of this degradation may be attributable to agricultural practices and other land-use pressures, climate change may also have altered the climatic conditions conducive to desertification and land degradation. We focus on this climatic dimension in our analysis. Specifically, interest lies in whether there has been a shift in the tail dependence structure of temperature and drought-related indicators across Spain. The IPCC Special Report on Climate Change and Land ([IPCC 2019](#)) highlights that the combination of drought and very high temperatures is generally more damaging than either stressor alone, for instance, when considering crop yields. Consequently, understanding whether such compound drought-heat events are becoming more frequent is crucial for policy planning.

Daily maximum temperature and precipitation data for 40 locations across Spain for 1960–2024 were obtained from the European Climate Assessment Dataset (ECAD) ([Klein Tank et al. 2002](#)). Figure 1 left panel shows that the sites considered cover all areas of Spain. We aggregate to a weekly resolution to reduce temporal dependence, taking the maximum recorded temperature and total accumulated precipitation within each week. Since land degradation is typically associated with prolonged drought rather than short-term precipitation deficits, we derive the Standardised Precipitation Index (SPI) [McKee, Doesken, and Kleist \(1993\)](#) from the time series of weekly precipitation totals using the SPEI R package ([Beguería & Vicente-Serrano 2023](#)).

The SPI measures deviations in accumulated precipitation from its long-term climatological distribution and therefore provides a more informative summary of drought conditions. Moreover, temperature dynamics may respond more strongly to drought conditions than the precipitation recorded during the current week. Formally, for a given accumulation period, the SPI is

computed by first aggregating precipitation over a rolling window and then fitting a gamma distribution to the historical series of accumulated precipitation totals. The cumulative probability associated with an observed accumulation is then transformed using the inverse cumulative distribution function of the standard normal distribution. In our application, the accumulation period was set to 13 weeks, including the current week. Since, under the usual SPI convention, negative values represent drier-than-normal conditions, we define the drought index $D = -\text{SPI}$. Larger values of D therefore correspond to more severe drought conditions. In what follows, we analyse the dependence between extremely high values for D and the weekly maximum temperature.

Figure 1 right panel suggests that the relationship between weekly maximum temperature and the drought index D is weaker for Daroca, Aragon, than for Jerez de la Frontera in Andalusia. These differences are broadly consistent with regional climatological patterns. Andalusia typically experiences a pronounced dry season during summer, with little precipitation over extended periods, whereas Aragon generally receives lower but more evenly distributed precipitation throughout the year. Consequently, drought conditions in Andalusia may be more closely linked to episodes of extreme heat than in Aragon. In what follows, we propose a methodology to determine whether there is a shift in the dependence of maximum weekly temperature and the drought index D for the most extreme events. This methodology will be applied to the data in Section 5. Due to slight seasonal differences in the dependence structure visible in Figure 1, we will analyse the seasons separately.

3 | METHODOLOGY

Suppose we have D spatial sites and observe d variables across T time points at each site; $D = 40$ and $d = 2$ in Section 2. Let $\tilde{\mathbf{X}}_s^{(t)} = (\tilde{X}_{1,s}^{(t)}, \dots, \tilde{X}_{d,s}^{(t)})^\top$ denote the vector of variables observed at site $s \in \{1, \dots, D\}$ and time $t \in \{1, \dots, T\}$, and we assume that $\tilde{\mathbf{X}}_s^{(t)}$ is independent of $\tilde{\mathbf{X}}_{s'}^{(t')}$ unless $(s, t) = (s', t')$. Our objective is to analyse the tail behaviour of $\tilde{\mathbf{X}}_s^{(t)}$, in particular the extremal dependence of the d variables at the individual sites across time. We want to further assess whether there is evidence of non-stationarity in the tail behaviour at one or more site. To perform inference, we assume that the tail dependence structure is stationary across subsets of the observation window. Let $\tau \in \{1, \dots, T\}$ be a time point and define the time blocks $\mathcal{T}_1(\tau) = \{\tau - n + 1, \dots, \tau\}$ and $\mathcal{T}_2(\tau) = \{\tau + 1, \dots, \tau + n\}$, where n is a user-specified block length. We then assume that the distribution of $\tilde{\mathbf{X}}_s^{(t)}$ remains unchanged across all time points within the same block. In the rest of the paper, we write $\mathbf{X}_j^{(i)}$ when referring to a random vector which has the same distribution as the random vectors across block $\mathcal{T}_j(\tau)$ and drop the index τ for notational brevity.

Subsection 3.1 summarises the conditional extremes (CE) framework by [Heffernan and Tawn \(2004\)](#) and [Keef, Papastathopoulos, and Tawn \(2013\)](#) which is used throughout and characterises the tail behaviour of $\mathbf{X}_s^{(j)}$ conditional on one of its components exceeding a high threshold. Section 3.2 outlines the divergence measure proposed by [O'Toole, Rohrbeck, and Richards \(2025\)](#), which we employ to quantify similarity of the extremal dependence for $\mathbf{X}_s^{(j)}$ and $\mathbf{X}_{s'}^{(j)}$ at two different sites $(s, s' = 1, \dots, D; j = 1, 2)$. Finally, Section 3.3 introduces our permutation-based procedure for detecting a structural breakpoint in tail behaviour using the pairwise similarities between fitted CE models.

3.1 | The conditional extremes model

Following [Keef, Papastathopoulos, and Tawn \(2013\)](#), we transform each component of $\mathbf{X}_s^{(j)}$ to standard Laplace margins, yielding the random vector $\mathbf{Y}_s^{(j)} = (Y_{1,s}^{(j)}, \dots, Y_{d,s}^{(j)})^\top$. Formally, we achieve this transformation by defining

$$Y_{i,s}^{(j)} = \begin{cases} \log \left\{ 2F_{i,s}^{(j)} \left(X_{i,s}^{(j)} \right) \right\} & \text{if } F_{i,s}^{(j)} \left(X_{i,s}^{(j)} \right) < 0.5, \\ -\log \left\{ 2 \left[1 - F_{i,s}^{(j)} \left(X_{i,s}^{(j)} \right) \right] \right\} & \text{if } F_{i,s}^{(j)} \left(X_{i,s}^{(j)} \right) \geq 0.5, \end{cases} \quad (1)$$

where $F_{i,s}^{(j)}(\cdot)$ is the cumulative distribution function of $X_{i,s}^{(j)}$ ($i = 1, \dots, d; s = 1, \dots, D; j = 1, 2$). Let $\mathbf{Y}_{-i,s}^{(j)}$ denote $\mathbf{Y}_s^{(j)}$ with its i^{th} component removed. The CE framework postulates that, for a sufficiently large threshold $u_i^{(j)}$, the conditional distribution of $\mathbf{Y}_{-i,s}^{(j)}$ given $Y_{i,s}^{(j)} = y$ can approximately be represented as

$$(\mathbf{Y}_{-i,s}^{(j)} \mid Y_{i,s}^{(j)} = y) = \boldsymbol{\alpha}_{|i,s}^{(j)} y + y^{\boldsymbol{\beta}_{|i,s}^{(j)}} \mathbf{Z}_{|i,s}^{(j)}, \quad y > u_i^{(j)}, \quad (2)$$

where $\alpha_{|i,s}^{(j)} \in [-1, 1]^{d-1}$, $\beta_{|i,s}^{(j)} \in (-\infty, 1]^{d-1}$, and all vector operations are applied componentwise. Further, the residual vector $Z_{|i,s}^{(j)}$ is assumed to be independent of $Y_{i,s}^{(j)}$. The parameter vectors $\alpha_{|i,s}^{(j)}$ and $\beta_{|i,s}^{(j)}$ govern, respectively, the strength of extremal dependence and the rate at which the conditional variability scales with the conditioning value.

For inference and to quantify similarity of CE models, we adopt the working assumption proposed by [Heffernan and Tawn \(2004\)](#) that

$$Z_{|i,s}^{(j)} \sim \text{MVN}_{d-1} \left(\mu_{|i,s}^{(j)}, \Sigma_{|i,s}^{(j)} \right),$$

which gives

$$(Y_{-i,s}^{(j)} | Y_{i,s}^{(j)} = y) \sim \text{MVN}_{d-1} \left(\alpha_{|i,s}^{(j)} y + y^{\beta_{|i,s}^{(j)}} \mu_{|i,s}^{(j)}, \Omega_{|i,s}^{(j)}(y) \right), \quad y > u_{i,s}^{(j)}, \quad (3)$$

where

$$\Omega_{|i,s}^{(j)}(y) = \text{diag} \left(y^{\beta_{|i,s}^{(j)}} \right) \Sigma_{|i,s}^{(j)} \text{diag} \left(y^{\beta_{|i,s}^{(j)}} \right).$$

We fit this model separately for each conditioning variable and spatial location using maximum likelihood estimation, without imposing spatial smoothness. The threshold $u_{i,s}^{(j)}$ is fixed at a high quantile u of the standard Laplace distribution, for all i and s , to ensure that the resulting CE models are comparable across locations.

Although the Gaussian assumption (3) is known to suffer from parameter identifiability issues and may lead to invalid inference ([Keef, Tawn, & Lamb 2013](#)), the results of [O'Toole et al. \(2025\)](#) suggest that these shortcomings have limited impact when the primary objective is to analyse spatial variation in extremal dependence among $Y_1^{(j)}, \dots, Y_D^{(j)}$. It is therefore reasonable to expect that the assumption remains adequate when additionally allowing for non-stationarity in the extremal dependence structure. Moreover, the assumption yields substantial computational advantages, as discussed in the next subsection.

3.2 | A divergence measure for conditional extremes models

[O'Toole et al. \(2025\)](#) introduce the conditional extremal skew-geometric Jensen–Shannon divergence as a dissimilarity measure between CE models; the measure is based on the skew-geometric Jensen–Shannon divergence [Nielsen \(2019\)](#). Let $H_{i,s}^{(j)}(y)$ denote the distribution function of the CE model in (3) for $Y_{-i,s}^{(j)} | (Y_{i,s}^{(j)} = y)$, $y > u$. The skew-geometric Jensen–Shannon divergence of $H_{i,s}^{(j)}(y)$ and $H_{i,s'}^{(j)}(y)$ is defined by

$$\text{JS}^{G_\lambda} \left[H_{i,s}^{(j)}(y) \| H_{i,s'}^{(j)}(y) \right] = \lambda \text{KL} \left\{ G_\lambda \left[H_{i,s}^{(j)}(y), H_{i,s'}^{(j)}(y) \right] \| H_{i,s}^{(j)}(y) \right\} + (1 - \lambda) \text{KL} \left\{ G_\lambda \left[H_{i,s}^{(j)}(y), H_{i,s'}^{(j)}(y) \right] \| H_{i,s'}^{(j)}(y) \right\}, \quad (4)$$

where KL denotes the Kullback–Leibler divergence, $G_\lambda \left[H_{i,s}^{(j)}(y), H_{i,s'}^{(j)}(y) \right]$ is the distribution with density proportional to the weighted geometric mean of the densities corresponding to $H_{i,s}^{(j)}(y)$ and $H_{i,s'}^{(j)}(y)$, and $\lambda \in [0, 1]$. We set $\lambda = 0.5$, yielding a symmetric, non-negative measure of dissimilarity. Due to the Gaussian assumption (3), the expression in (4) can be evaluated in closed form; see [Nielsen \(2019\)](#) and [O'Toole et al. \(2025\)](#) for details.

The quantity (4) only compares the fitted CE models at the two locations for $Y_{i,s}^{(j)} = Y_{i,s'}^{(j)} = y$. We therefore take its expectation with respect to the conditioning variable. As all marginal distributions are standard Laplace and the conditioning threshold is fixed as u across locations (see Section 3.1), the expected skew-geometric Jensen–Shannon divergence for two CE models is

$$E_{i,s,s'}^{(j)} = \mathbb{E} \left\{ \text{JS}^{G_\lambda} \left[H_{i,s}^{(j)}(Y) \| H_{i,s'}^{(j)}(Y) \right] | Y > u \right\} = \int_u^\infty \text{JS}^{G_\lambda} \left[H_{i,s}^{(j)}(y) \| H_{i,s'}^{(j)}(y) \right] f(y - u) dy, \quad (5)$$

where $f(y - u) = \exp\{-(y - u)\}$ is the density of $Y_{i,s} | (Y_{i,s} > u)$ for $u > 0$; we refer the reader to [O'Toole et al. \(2025\)](#) for a theoretical treatment of this dissimilarity measure. In practice, we approximate the expectation using Monte Carlo integration and, to avoid unstable extrapolation beyond the observed range, we evaluate the integral only up to a suitably high upper quantile of the standard exponential distribution. By averaging $E_{i,s,s'}^{(j)}$ over the d conditioning variables for each pair (s, s') of sites, we obtain the dissimilarity matrix $M^{(j)} \in \mathbb{R}^{D \times D}$, with

$$M^{(j)} = \left\{ M_{s,s'}^{(j)} \right\}_{s,s'=1,\dots,D}, \quad M_{s,s'}^{(j)} = \frac{1}{d} \sum_{i=1}^d E_{i,s,s'}^{(j)}, \quad t = 1, \dots, T. \quad (6)$$

The resulting matrix forms the basis for the procedure described in the next subsection for analysing non-stationarity in the tail behaviour.

3.3 | Testing for non-stationarity in extremal dependence

To assess stationarity in the tail behaviour of the D random vectors around time $\tau \in \{1, \dots, T\}$, we construct a statistical test based on the pairwise dissimilarities of CE models fitted using observations from the two time blocks $\mathcal{T}_1(\tau)$ and $\mathcal{T}_2(\tau)$. Under stationarity, we would have $M^{(1)}(\tau) = M^{(2)}(\tau)$ for the dissimilarities matrix in (6). This motivates the null hypothesis

$$H_0 : M^{(1)}(\tau) = M^{(2)}(\tau), \quad (7)$$

with the alternative hypothesis $H_1 : M^{(1)}(\tau) \neq M^{(2)}(\tau)$.

Given the data for the time blocks $\mathcal{T}_1(\tau)$ and $\mathcal{T}_2(\tau)$, we obtain estimates $\hat{M}_1(\tau)$ and $\hat{M}_2(\tau)$ of $M^{(1)}(\tau)$ and $M^{(2)}(\tau)$, respectively, by first estimating the CE model parameters via maximum likelihood inference, followed by Monte Carlo integration for evaluating (5). Two popular statistics are considered to test H_0 against H_1 . The first is the Frobenius norm of the difference between $\hat{M}^{(1)}(\tau)$ and $\hat{M}^{(2)}(\tau)$:

$$N_{\text{obs}}^{(F)}(\tau) := \left\| \hat{M}^{(1)}(\tau) - \hat{M}^{(2)}(\tau) \right\|_F = \sqrt{\sum_{s=1}^D \sum_{s'=1}^D \left[\hat{M}_{s,s'}^{(1)}(\tau) - \hat{M}_{s,s'}^{(2)}(\tau) \right]^2}. \quad (8)$$

The second statistic is the infinity, or maximum-entry, norm:

$$N_{\text{obs}}^{(I)}(\tau) := \max_{s=1, \dots, D} \sum_{s'=1}^D \left| \hat{M}_{s,s'}^{(1)}(\tau) - \hat{M}_{s,s'}^{(2)}(\tau) \right|. \quad (9)$$

The Frobenius norm (8) captures the overall magnitude of change, whereas the infinity norm (9) is only sensitive to the largest pairwise difference. Consequently, their relative performance under non-stationarity is expected to depend on whether changes occur across many sites or are concentrated in a small number of locations. For notational brevity, we write $N_{\text{obs}}(\tau)$ in the rest of the section when referring to $N_{\text{obs}}^{(F)}(\tau)$ or $N_{\text{obs}}^{(I)}(\tau)$. We recognise that other statistics for the matrix difference could be considered, but we focus on these two for their popularity, simplicity and interpretability, as an illustration of the proposed permutation testing procedure.

To determine whether to reject H_0 based on $N_{\text{obs}}(\tau)$, we implement a permutation test, which involves comparing the observed test statistic to a reference distribution generated by randomly permuting the data (Good 2004). This approach avoids strong parametric assumptions about the sampling distribution of the dissimilarity matrices and is exact when the observations are exchangeable under H_0 . The null distribution of $N_{\text{obs}}(\tau)$ is approximated using B permutations:

1. Generate B random permutations of the $2n$ observations from the period $\mathcal{T}_1(\tau) \cup \mathcal{T}_2(\tau)$. For each permutation, assign the first n observations to one block, and the remaining n observations to the second block.
2. For each permutation $b \in \{1, \dots, B\}$, estimate the CE model parameters and, subsequently, the matrix estimates, yielding estimates $\hat{M}_b^{(1)}(\tau)$ and $\hat{M}_b^{(2)}(\tau)$.
3. Compute the test statistic N_b as defined in (8) or (9) using $\hat{M}_b^{(1)}(\tau)$ and $\hat{M}_b^{(2)}(\tau)$.

The empirical p-value for the test is then

$$\hat{p} = \frac{1 + \sum_{b=1}^B \mathbf{1}\{N_b \geq N_{\text{obs}}(\tau)\}}{B + 1}, \quad (10)$$

where $\mathbf{1}\{\cdot\}$ denotes the indicator function. A value of $\hat{p}$ below the significance level α , $\hat{p} < \alpha$, provides evidence for a difference in the pairwise dissimilarity structure of the fitted CE models between the two blocks, suggesting non-stationarity in tail behaviour around time point τ .

When the number of values considered for τ is modest, as in the simulation study and Spanish application in Sections 4 and 5, permutation testing can be performed over the complete candidate range. In settings with longer time series or a substantially larger number of sites, one may first identify local maxima, thereby reducing computational cost. Moreover, a suitable block size n must be chosen to ensure reliable estimation of the CE model parameters within each time block while providing sufficient power to detect changes. We apply the same n at every candidate τ within an analysis and consider only time points satisfying $\tau \in \{n, \dots, T-n\}$. Finally, the proposed procedure assesses non-stationarity locally around a given τ . To test for non-stationarity across the entire observation period, one may either test every admissible τ with an appropriate multiple-testing correction or define a global test statistic

$$N_{\text{max}} := \max_{\tau=n, \dots, T-n} N_{\text{obs}}(\tau).$$

A permutation test can then be constructed using the corresponding maximum statistic under each permutation. This automatically accounts for the multiple testing induced by considering multiple values of τ .

4 | SIMULATION STUDY

We demonstrate the ability of the proposed procedure to detect temporal changes in multivariate extremal dependence through a simulation study. The simulation design follows the observational structure of the Spanish drought data introduced in Section 2, with two variables observed weekly at multiple sites and grouped into season-years of 12 observations. Rather than imposing an instantaneous changepoint, we allow the extremal dependence structure to evolve gradually over a specified transition period, to better reflect the types of changes that may occur in environmental systems. The objective is therefore not to recover a single exact change time, but to assess whether the procedure identifies evidence of a transition and whether the Frobenius and infinity norms respond differently to widespread and localised changes.

In Section 4.1, we describe the simulation setup and the scenarios considered, which include a null scenario with no change, a local change affecting a subset of sites, and a global change affecting all sites. Section 4.2 presents the results of the simulation study, comparing the performance of the Frobenius and infinity norms in detecting changes under each scenario.

4.1 | Simulation setup

Rewrite to reflect new setup

Following Section 2, we simulate two variables at $D = 40$ sites over 60 season-years, with 12 weekly observations in each season-year. At each site and time point, the two variables are generated from a bivariate Student t -copula with three degrees of freedom (Demarta & McNeil 2007). The sites are generated independently, and the weekly observations are conditionally independent given the site- and year-specific copula parameters. The observations are transformed to standard Laplace margins using their known marginal distributions. Across all CE fits, the same threshold u , set to the 85th quantile of the standard Laplace distribution, as well as 500 Monte Carlo samples of the conditioning variable, are used to evaluate the expected divergence in Equation (5), to ensure comparability of the resulting conditional distributions.

The sites are divided into three baseline extremal dependence groups. Sites 1, ..., 14 have initial t -copula correlation parameter $\rho_{s,0} = 0.1$, sites 15, ..., 27 have $\rho_{s,0} = 0.5$, and sites 28, ..., 40 have $\rho_{s,0} = 0.8$. These groups produce heterogeneous extremal dependence across the spatial domain before any temporal change is introduced.

For an affected site s , we allow the t -copula correlation parameter to evolve according to

$$\rho_{s,t} := \tanh \left\{ \operatorname{atanh}(\rho_{s,0}) + A_s \delta_z g(t) \right\}, \quad (11)$$

where A_s indicates whether site s is affected by the change, δ_z controls its magnitude on the Fisher-z scale, and $g(t)$ determines its temporal progression. Specifically,

$$g(t) = \begin{cases} 0, & t \leq t_0, \\ \frac{t - t_0}{t_1 - t_0}, & t_0 < t < t_1, \\ 1, & t \geq t_1. \end{cases} \quad (12)$$

so that the dependence parameter is constant before season-year t_0 , changes gradually between t_0 and t_1 , and remains constant thereafter. The transformations $\operatorname{atanh}(\cdot)$ and $\tanh(\cdot)$ allow the change to be specified using an unrestricted linear predictor while ensuring that $\rho_{s,t} \in (-1, 1)$ throughout.

We consider three scenarios:

- (i) **Null:** no sites are affected, so that $A_s = 0$ for all s ;
- (ii) **Local change:** five sites from the first baseline group are affected, so that $A_s = 1$ for $s = 1, \dots, 5$ and $A_s = 0$ otherwise;
- (iii) **Global change:** all 40 sites are affected, so that $A_s = 1$ for every s .

For the global case, we set $\delta_z = 0.45$, under which an affected site with baseline correlation $\rho_{s,0} = 0.1$ reaches a final correlation of approximately 0.5. For the other two baseline groups, the corresponding final correlations are approximately 0.76 and 0.91.

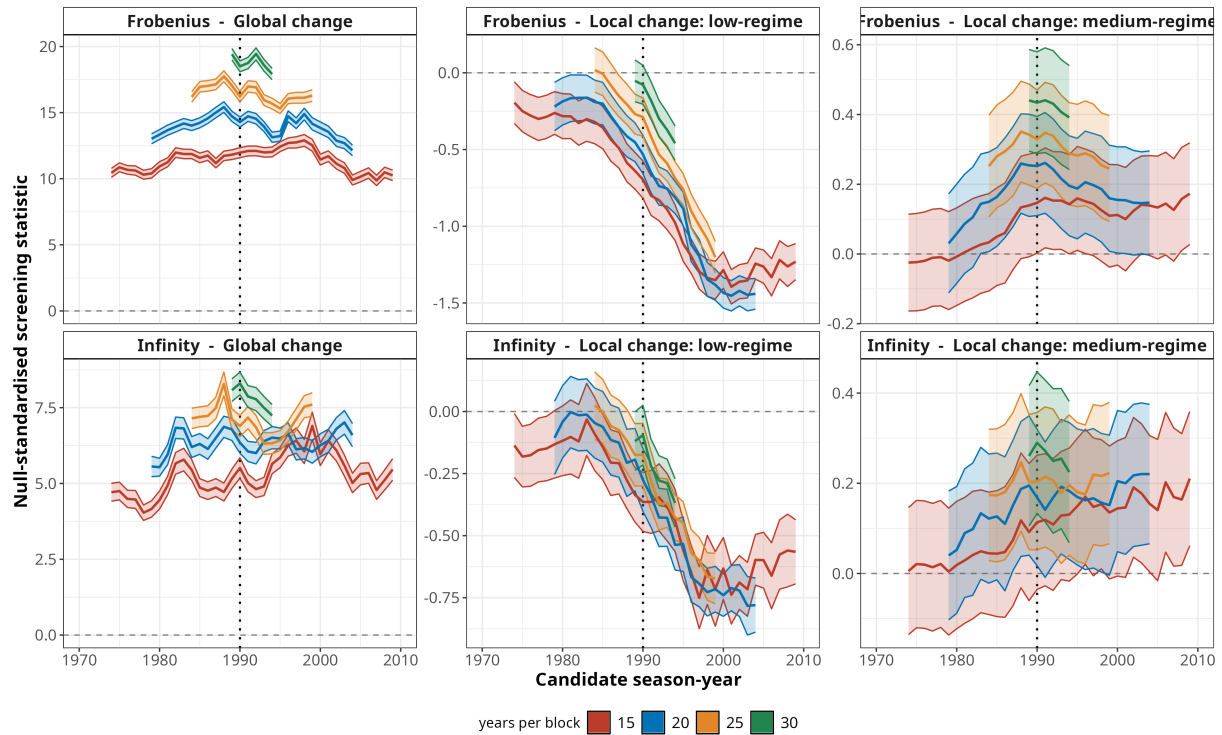


FIGURE 2 ???

For the local case, we set $\delta_z = 0.21$ and $\delta_z = 0.59$, so that the affected sites reach correlations of approximately 0.3 and 0.6, respectively. The local scenario represents a concentrated change in the extremal dependence structure, whereas the global scenario represents a broader system-wide transition. We therefore expect the infinity norm to be particularly responsive to the local alternative and the Frobenius norm to benefit from the accumulation of pairwise changes under the global alternative.

Say about value of B (number of iterations)

For each scenario, we generate 100 independent datasets and apply the changepoint procedure described in Section 3.3. The CE models are fitted using adjacent blocks of 25 season-years on either side of each candidate, and candidate split points are evaluated over the complete feasible range. This block length retains a useful range of candidate locations while providing sufficiently many observations for stable tail estimation. For each replicate and norm, we record the screening statistic $D(\tau)$ and the corresponding permutation p -values $\hat{p}(\tau)$ for each candidate split point τ , for $B = 100$ permutations. Under the null scenario, the p -values give an empirical frequency with which the procedure identifies an apparent change in a temporally stationary dependence structure. Under the local and global alternatives, they measure the ability of the procedure to detect the simulated transition and identify the region to which the estimated change is localised.

4.2 | Simulation results

Insert screening step section analysis here

- At each candidate split point, the pointwise rejection rate is the proportion of simulation replicates for which the permutation p -value is below the specified significance level.
- Under the null scenario, this estimates the pointwise Type I error rate; under either alternative, it estimates the pointwise power.
- The uncertainty intervals shown below are 95% Wilson confidence intervals for these binomial proportions. item They quantify Monte Carlo uncertainty due to the finite number of simulation replicates.

Not the case at all!

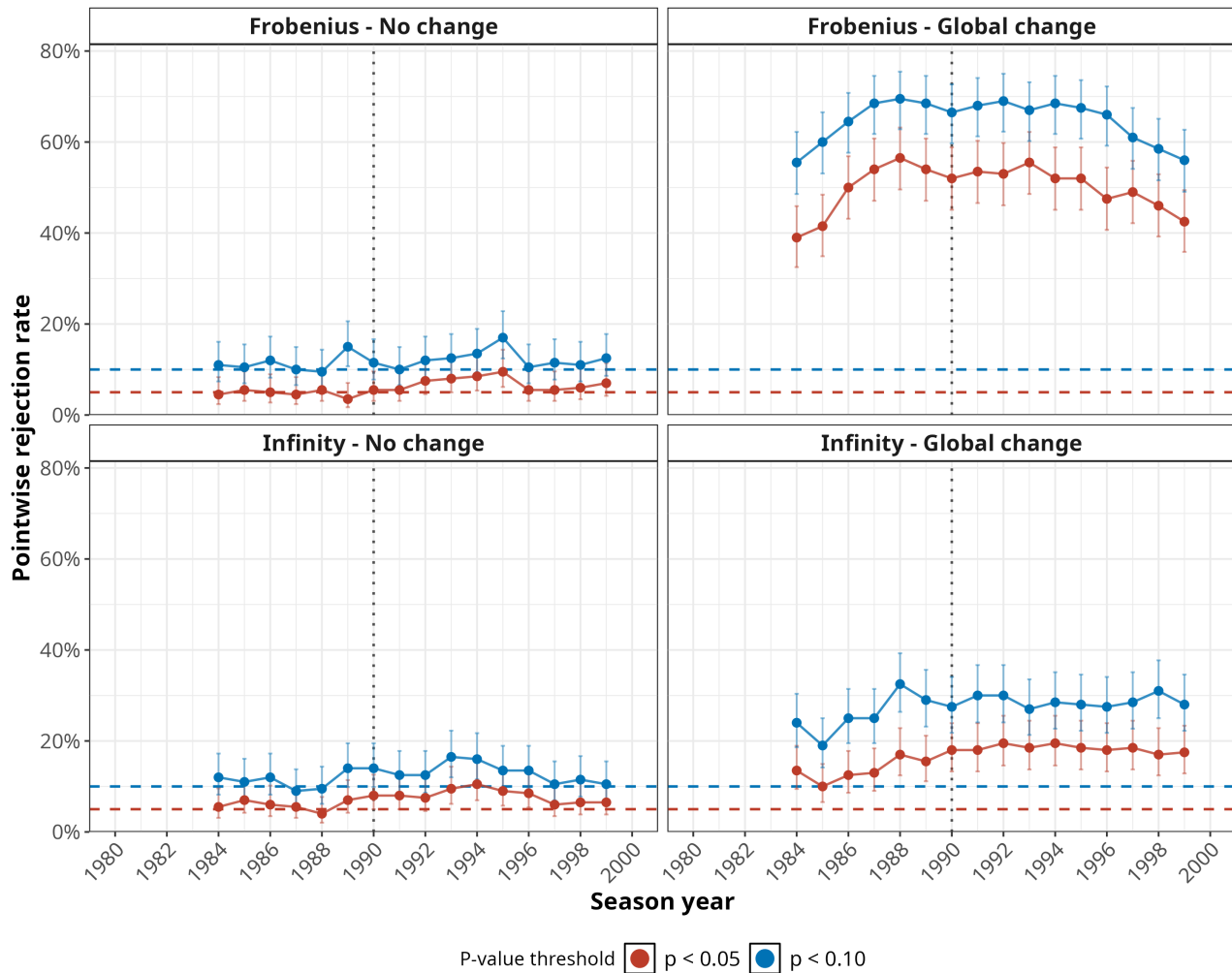


FIGURE 3 Pointwise rejection rates under the null and global-change scenarios for the Frobenius and infinity norms. Red and blue curves correspond to significance levels of 5% and 10%, respectively, and the horizontal dashed lines indicate the corresponding nominal levels. The vertical dotted line marks the onset of the gradual change in 1990; in the null panels, it is included only as a reference. Points show the proportion of the 100 simulation replicates rejected at each candidate split point, and vertical bars give 95% Wilson confidence intervals.

- Figure 3 compares the pointwise Type I error rates under the null scenario with the pointwise power under the global-change scenario.
- Under the null scenario, the rejection rates generally remain close to their nominal levels.
- For the Frobenius norm, the rejection rates range approximately from 2% to 8% at the 5% level and from 7% to 13% at the 10% level.
- For the infinity norm, the corresponding ranges are approximately 1%–9% and 8%–14%.
- The Wilson confidence intervals generally contain the nominal significance levels.
- The observed variation between candidate split points is therefore compatible with the Monte Carlo uncertainty associated with 100 simulation replicates, suggesting that both statistics are reasonably well calibrated pointwise.

Change 1995 to 1999

- Under the global-change scenario, the rejection rates are consistently higher than under the null, particularly for the Frobenius norm.
- The increase is most apparent for candidate split points at and after the onset of the gradual transition.
- At the 5% level, the rejection rate for the Frobenius norm increases from approximately 8% at the 1985 candidate to 16% at 1990 and 32% at 1995.
- At the 10% level, the corresponding rates are approximately 16%, 34% and 40%.
- By comparison, the null rejection rates at the 1995 candidate are approximately 8% and 13% at the 5% and 10% levels, respectively.
- The late-candidate rejection rates under the global alternative therefore represent a clear increase relative to the empirical null baseline.
- The infinity norm also exhibits increased power under the global alternative, but the improvement is less pronounced.
- At the 1995 candidate, its rejection rates are approximately 22% at the 5% level and 28% at the 10% level, compared with null rejection rates of approximately 9% and 14%, respectively.
- The stronger performance of the Frobenius norm is consistent with the global nature of the alternative.
- As the dependence structure changes at every site, differences accumulate across many entries of the dissimilarity matrix and consequently contribute to the Frobenius norm.
- The infinity norm instead depends only on the largest entrywise difference.
- The power profiles are neither sharply peaked nor strictly monotone.
- This is consistent with the gradual nature of the simulated transition: there is no unique instantaneous changepoint, and nearby candidate splits compare temporal blocks containing different mixtures of pre-transition and transitional observations.
- Even under the global alternative, the largest observed pointwise power is approximately 40%.
- The procedure therefore has moderate rather than high power under this simulation design.
- Figure 4 compares the empirical null rejection rates with the pointwise power obtained when the gradual change is restricted to five of the 40 sites.
- When the correlation parameter at the affected sites increases from 0.1 towards 0.3, the rejection profiles for both norms remain similar to those obtained under the null scenario. This magnitude of local change is therefore not reliably detectable under the present simulation design.
- Increasing the final correlation to 0.6 produces a clearer response for the Frobenius norm. At the 10% significance level, its rejection rate increases from approximately 16% at the 1990 candidate to 23% at 1995, compared with null rates of approximately 12% and 13%, respectively.
- At the 5% level, the rejection rate for the Frobenius norm reaches approximately 12%–13% for some of the later candidate splits, compared with approximately 5%–8% under the null.
- The improvement in power is therefore present but modest.
- The Wilson intervals overlap substantially across the null and local-change scenarios.
- Consequently, the evidence for improved detection under the stronger local alternative is better characterised by the overall increase across the later candidate splits than by any individual candidate year.

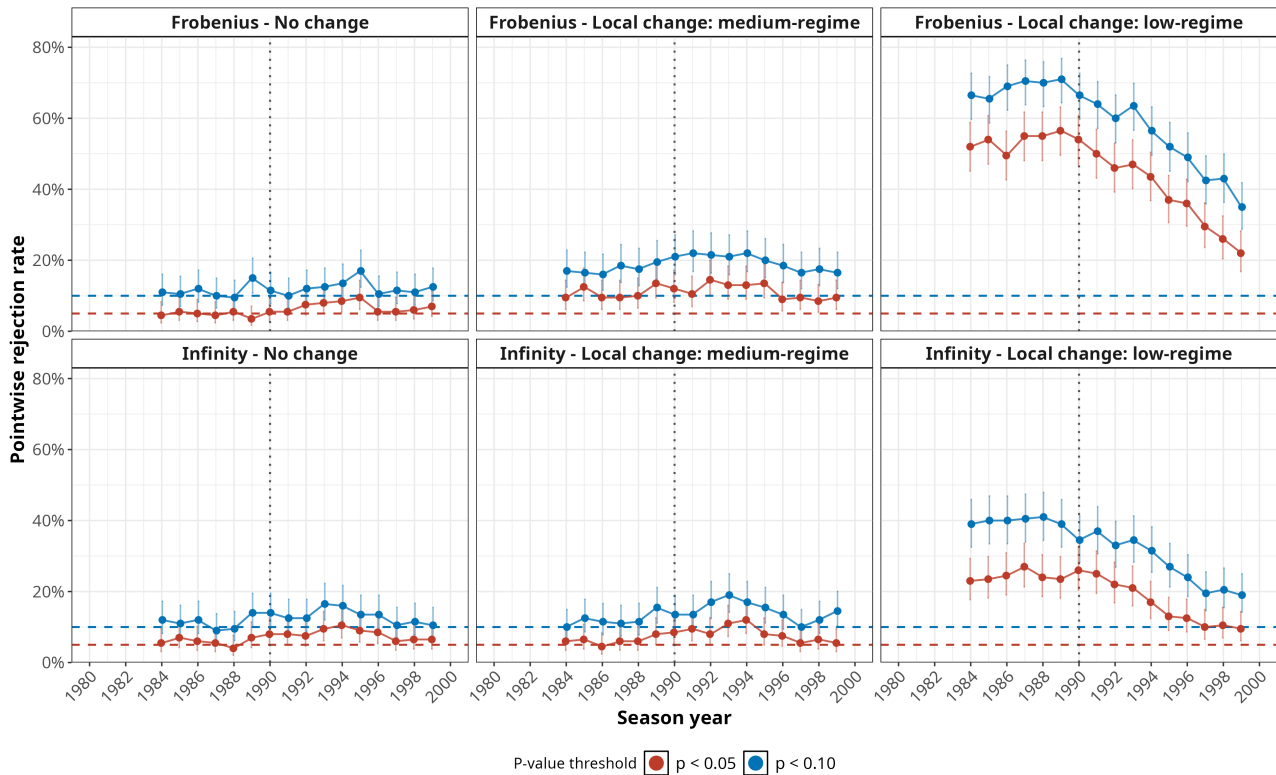


FIGURE 4 Pointwise rejection rates under the null scenario and the two local-change scenarios for the Frobenius and infinity norms. The local alternatives increase the correlation parameter at the five affected sites from 0.1 towards 0.3 or 0.6. Red and blue curves correspond to significance levels of 5% and 10%, respectively, and the horizontal dashed lines indicate the corresponding nominal levels. The vertical dotted line marks the onset of the gradual change in 1990; in the null panels, it is included only as a reference. Points show the proportion of the 100 simulation replicates rejected at each candidate split point, and vertical bars give 95% Wilson confidence intervals.

- The infinity norm shows little systematic improvement under either local alternative.
- Under the stronger change, its rejection rate at the 1995 candidate is approximately 9% at the 5% level and 17% at the 10% level, compared with null rates of approximately 9% and 14%, respectively.
- Thus, contrary to the initial expectation, the infinity norm does not provide greater sensitivity to the localised change considered here.
- One possible explanation is that the gradual change does not produce an individual entrywise difference sufficiently large to dominate the estimation variability in the dissimilarity matrices, instead producing differences in entire rows and columns of the matrix, which is more effectively captured by the Frobenius norm.
- Overall, the procedure is reasonably well calibrated pointwise and has greater power against the global alternative than against either local alternative.
- Detectability depends on both the magnitude and spatial extent of the change, as well as on whether it produces an accumulated signal or a sufficiently large entrywise difference in the estimated dissimilarity matrix.

5 | APPLICATION TO SPANISH DROUGHT AND TEMPERATURE DATA

5.1 | Testing for a shift in tail dependence structure

We organise the observations into *season-years*, with 12 weekly observations per season. For spring, summer and autumn, the season-year is the calendar year in which the observations occur, while December is assigned to the same season-year as the following January and February for the winter season. The observations per season are then grouped into separate datasets, resulting in four datasets, one per season. In what follows, the framework in Section 3, including the marginal standardisation, is applied to each dataset individually. When performing the permutation test later, observations in the same season-year form a single unit that is not split across permutations, and the candidate time τ refers to a year instead of a specific date. By permuting complete season-years, we preserve both the dependence between observations within a season-year and the spatial correspondence between stations.

The time series of maximum weekly temperature and drought index exhibit seasonality and possibly a long-term trend due to climate change. Since we assume that the marginal distribution function in (1) is stationary across the time block, changes in the marginal distributions over time may confound changes in the tail dependence if not modelled directly. Therefore, for each of the four datasets, we fit a linear trend for the average maximum weekly temperature for each month and station separately, which allows for a varying magnitude of temporal trend across space and time. After removing the trend, observations are transformed to Laplace margins using (1), with $F_{i,s}^{(j)}$ being set to the empirical cumulative distribution function of all observations in the same season. **CR: Is there an issue that variance may be different across months too and we have not account for it?**

To apply the framework in Section 3, we select the block length n such that each time block contains 25 season-years and only consider τ within the period 1985 – 1998, which ensures a sufficient number of data within each block. Sensitivity to the choice of n was explored, but a block size of 15 years or smaller was found to include too few exceedances to achieve a reliable fit of the CE models. For each τ and season, parameters of the CE model (2) are estimated separately for each site using the data within the time blocks $\mathcal{T}_1(\tau)$ and $\mathcal{T}_2(\tau)$. The threshold $u_{i,s}^{(j)}$ is set to the 80th percentile of the standard Laplace distribution, because it provides a good compromise between restricting attention to the joint tail while retaining sufficient exceedances for inference. For each of the 24 candidate years, we derive the test statistics $N_{obs}^{(F)}(\tau)$ and $N_{obs}^{(I)}(\tau)$ defined in Equations (8) and (9). **How to justify choice of 25 season-years without forward referencing to application?**

Figure 5 shows different patterns in the test statistics for the four seasons. For the winter months, the highest value is obtained for 1998 for both test statistics, while for the other seasons the highest test statistic is observed between 1988 and 1992. We also note a difference in scale, with the test statistic being smaller in spring than in the other seasons. Consequently, there appears to be a stronger signal for a shift in tail dependence for summer, autumn and winter than for spring. We then compute empirical p-values for the observed test statistics using the permutation procedure described in Section 3.3, with the number of permutations in (10) set to $B = 200$, and all conclusions being made at the 5% significance level. Figure 6 suggests that there is evidence of a significant difference for winter when τ is around 1998, and for summer for $\tau \in \{1987, \dots, 1993\}$. Conversely, there is insufficient evidence for a change for spring and autumn, which is in line with the highest test statistic for the seasons illustrated in Figure 5. Moreover, the conclusions are similar for the two test statistics used, with the infinity norm producing smaller p-values when the null hypothesis of no change is rejected at the 5% significance level.

- In particular, the resulting clustering solutions exhibited clear spatial contiguity. **Add a sensitivity analysis for the dependence threshold.**
- The Monte Carlo approximation used to construct the pairwise dissimilarities employs the same sample from the standard Laplace distribution in every temporal comparison. Holding this sample fixed avoids introducing additional Monte Carlo variation between candidate changepoints. **CR: This may be too niche a comment to make here.**
- The conditional fits from the two conditioning directions are combined into the aggregated dissimilarity matrix defined in Section 3. **Test for others**
- All screening, testing and clustering calculations reported below use this matrix. **Confirm terminology and add sensitivity results for the alternative dissimilarity matrices, if required.**
- Twenty-year blocks were reliable for winter but produced substantial failure rates for several other seasons.
- Also for Autumn, none of the three norms produces a pointwise p-value below 0.05, although the infinity norm came very close for 1986, the year with the highest test statistic in our screening step. **May have to rewrite slightly, as Autumn now**

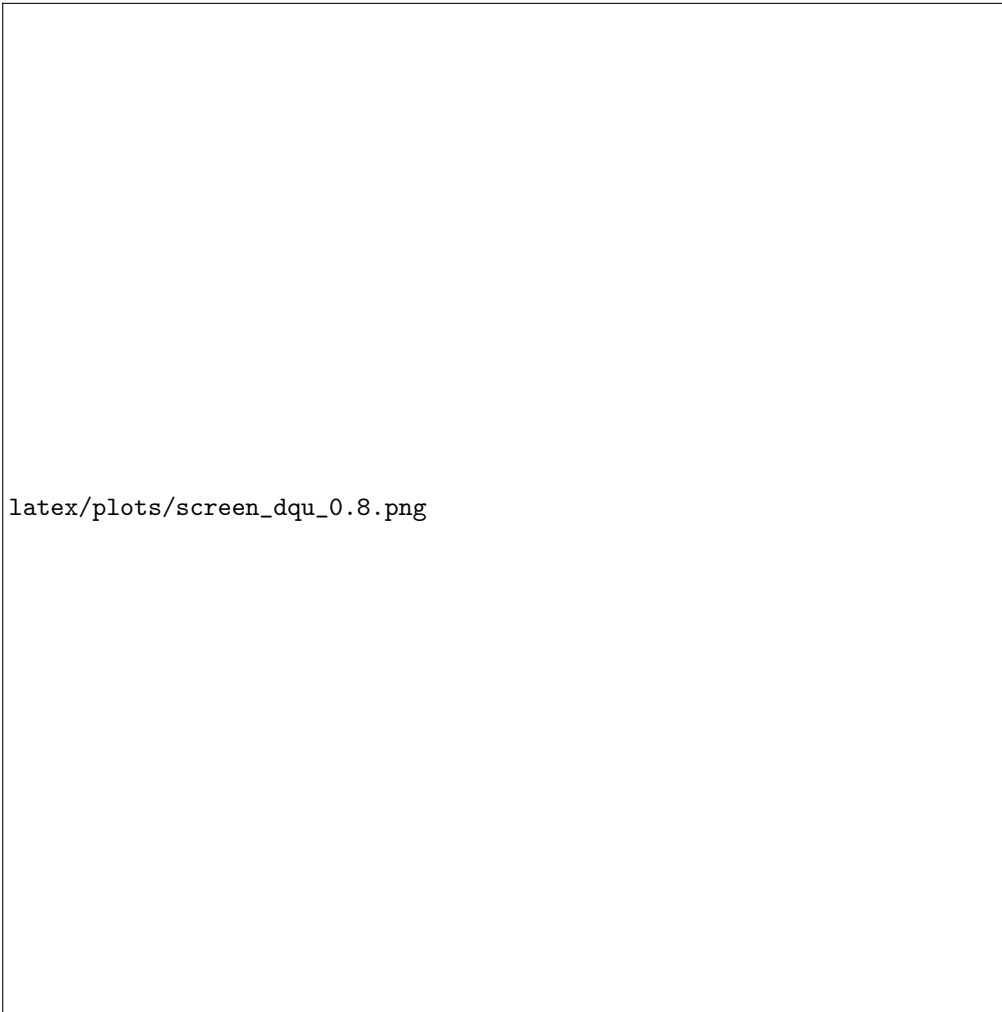


FIGURE 5 Test statistics based on the Frobenius (left) and infinity (right) norm for each candidate season-year based on 25-year time blocks. Local maximum are highlighted by the crosses, and test statistics have been transformed to a common scale. Check that standardising to common scale is correct

has a 5% changepoint (even here, it's significant at the 10% level) CR: I would say we stick with the 5% level, because it is consistent with our cluster analysis.

5.2 | Clustering Results

Are elbow plots briefly alluded to in methods?

- We finally investigate how the detected temporal changes are reflected in the spatial clustering of the fitted extremal dependence structures.
- For each fitted conditional extremes model, the aggregated dissimilarity matrix is used to partition the 40 stations into three clusters using the procedure described in Section 3.
- For summer, the pointwise tests identify a broader region of evidence during the late 1980s and early 1990s. We use 1990 as a representative division and fit separate models to season-years 1960–1990 and 1991–2024.
- For winter, the clearest common evidence across the three matrix norms occurs at season-year 1998. Separate models are therefore fitted to season-years 1960–1998 and 1999–2024.

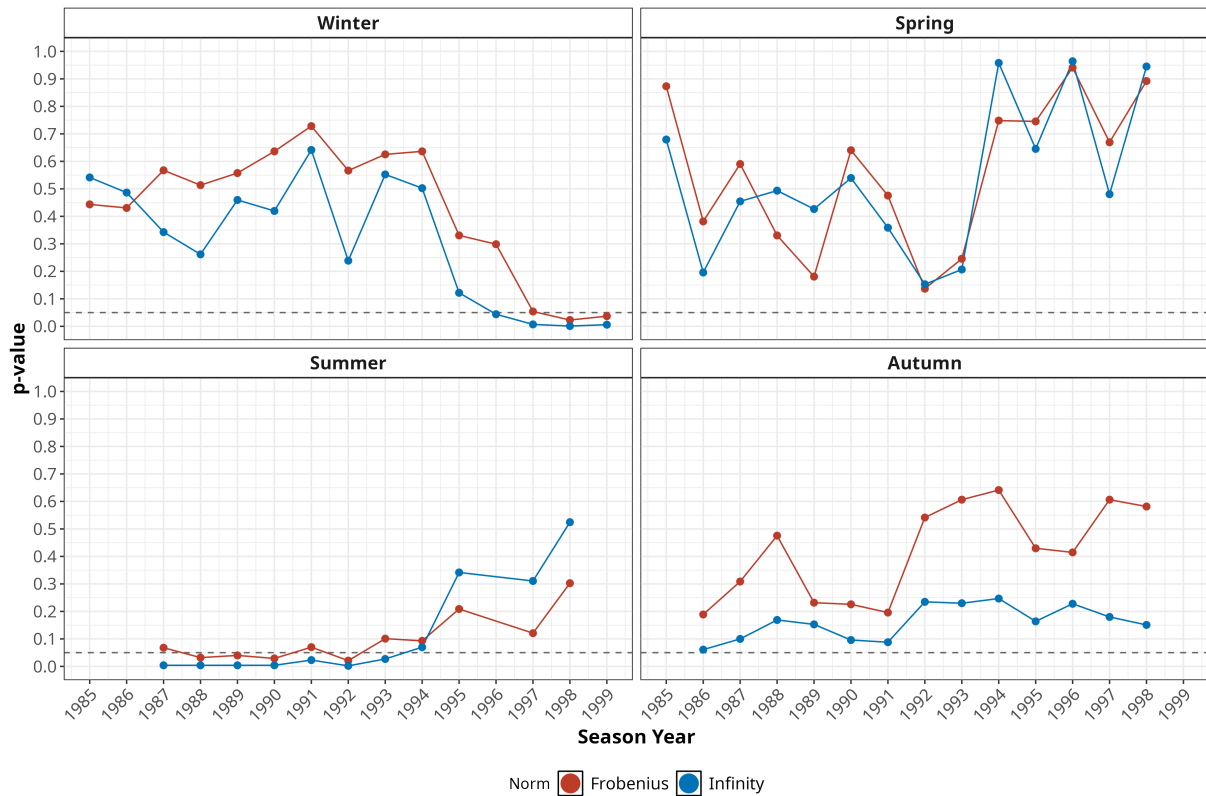


FIGURE 6 Empirical p-values for the different candidate years τ for the test statistics based on the Frobenius (red) and infinity (blue) norm. Estimates are based on 25 season-years per temporal block and $B = 200$ permutations. The horizontal dashed line corresponds to the pointwise 5% significance level.

- For autumn, the Infinity norm identifies possible evidence of a change at 1986 at the 10% pointwise significance level. We consequently examine the periods 1960–1986 and 1987–2024 as an exploratory sensitivity analysis.
- These divisions treat a candidate (t) as a change between season-years (t) and ($t + 1$).
- The changepoint analysis provides no consistent evidence of temporal change in spring. We therefore fit a single conditional extremes model to the complete spring record.
- The total within-cluster dissimilarity curves for each season and their respective divisions are shown in Figure 7 .
- Together, they generally support $k = 3$ as an interpretable compromise between parsimony and within-cluster homogeneity.
- However, as $k = 2$ and $k = 4$ also appear to be plausible for some elbow plots, we provide the corresponding clustering solutions in the Supplementary Material (Figures S7– S9).
- Figure 8 presents the resulting spatial clustering solutions for each season with a detected changepoint.
- The clusters exhibit substantial spatial coherence, despite station coordinates, elevation and other geographical covariates not entering the clustering procedure directly.
- No individual geographical covariate completely determines cluster membership. Nevertheless, longitude, latitude and distance to the coast exhibit clearer associations with membership than elevation, suggesting that the estimated extremal dependence structures contain a broad geographical component.

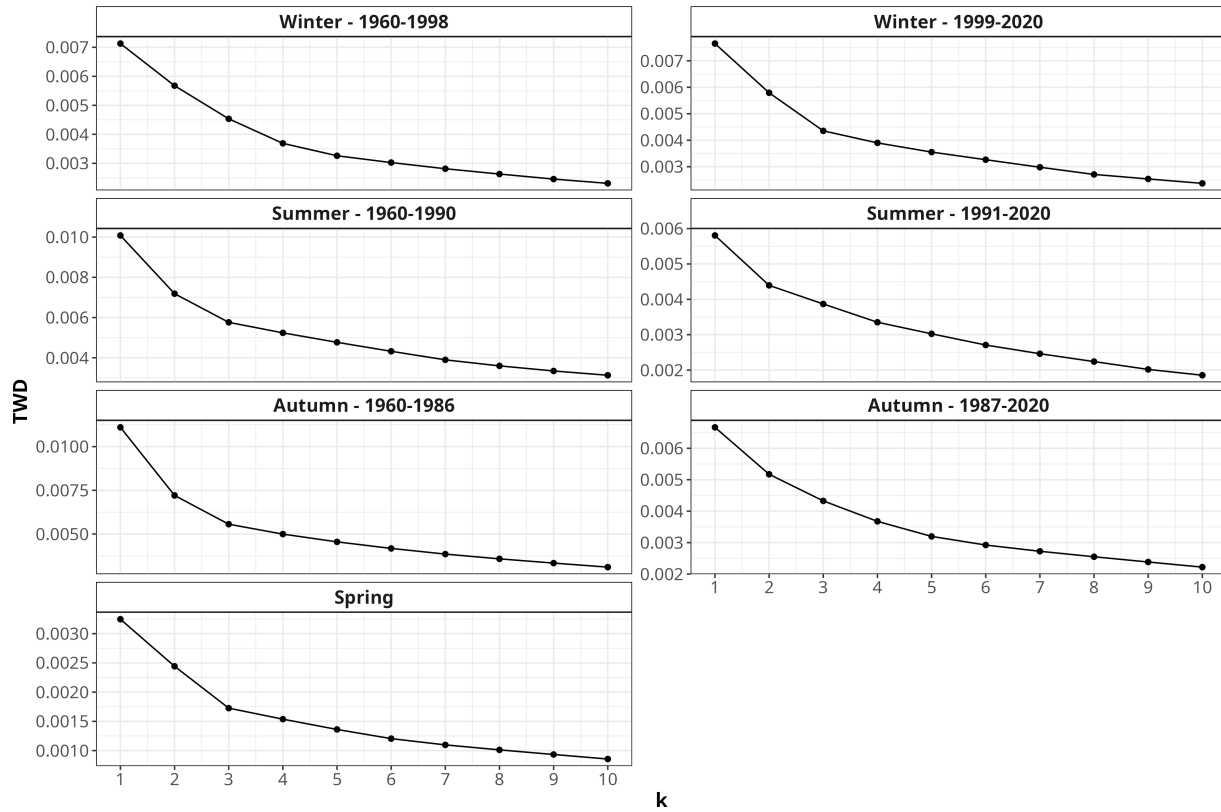


FIGURE 7 “Elbow” plots of the total within-group dissimilarity (TWD) against the number of clusters k for the estimated clusters, obtained using the aggregated dissimilarity matrix for each season and their respective changepoint divisions.

- Several geographically separated stations are assigned to the same cluster. The clustering therefore does not merely recover contiguous geographical regions, but identifies locations with similar fitted conditional extremal dependence structures.
- Appreciable differences are observed between the pre- and post-division solutions for winter, summer and autumn, although the spatial extent and direction of the reorganisation differ between seasons.
- For winter during 1960–1998, cluster 2 is concentrated principally along the southern and southeastern coast, with one geographically separated member on the northern coast. Cluster 3 is concentrated more strongly across northern and northwestern Spain, while cluster 1 contains most of the remaining stations.
- The early-winter solution is therefore spatially coherent, although geography is not a definitive determinant of membership.
- The corresponding dissimilarity matrix (Supplementary Figure S2) shows comparatively distinct groups, with cluster 3 displaying particularly large dissimilarities from the other two clusters.
- The elbow for 1960–1998 in Figure 7 is less sharply defined than for the later period and indicates that $k = 4$ is also plausible. Under $k = 4$, the original cluster 1 is subdivided into two larger groups, while only a small number of stations remain geographically separated from the other members of their assigned cluster.
- This sensitivity analysis suggests that the pre-1999 winter dependence structure may contain finer geographical variation than is represented by the three-cluster solution. It does not, however, provide unequivocal evidence that four clusters are optimal.
- The clustering changes appreciably after 1998, although some broad spatial structure remains. Cluster 1, which dominates the earlier solution, is substantially smaller during 1999–2024, while cluster 2 extends across a wider range of northern and interior locations.

- The post-1998 solution appears less geographically delineated, with a greater number of neighbouring stations assigned to different clusters and several clusters spanning geographically separated regions.
- The elbow curve for 1999–2024 provides clearer support for three clusters. Nevertheless, the corresponding dissimilarity matrix appears less block-structured and contains generally smaller between-station dissimilarities. **Support this statement by reporting the mean, median and maximum off-diagonal dissimilarities for each period.**

- The signed matrix difference (Supplementary Figure S6) is defined as

$$\Delta M^{(W)} = M_{1960-1998}^{(W)} - M_{1999-2024}^{(W)}$$

Positive entries therefore indicate station pairs whose fitted extremal dependence structures became more similar after 1998, whereas negative entries indicate pairs that became less similar.

- The winter difference matrix contains both positive and negative entries. The temporal change is therefore heterogeneous rather than a uniform convergence across all stations.
- Prominent positive bands are associated with Murcia and Ponferrada, indicating that their fitted dependence structures became more similar to those at several other stations after 1998.
- Both stations also change cluster membership: Murcia moves from cluster 2 to cluster 3, while Ponferrada moves from cluster 3 to cluster 2. These transitions illustrate that a reduction in pairwise dissimilarity does not necessarily preserve the original cluster boundaries.
- The broad spatial reorganisation in winter is consistent with the changepoint signal being visible under both the Frobenius and Infinity norms: the difference matrix contains several prominent station-specific changes together with more widely distributed moderate changes.
- Overall, the winter results provide evidence that the geographical separation of the fitted extremal dependence regimes weakened after 1998. This should be interpreted as convergence among fitted dependence structures, rather than as evidence that the absolute strength of extremal dependence increased.
- The full-record spring solution (Supplementary Figure S1) also exhibits substantial spatial coherence.
- Cluster 2 extends along much of the northern and northwestern coast, whereas clusters 1 and 3 occupy broader and partially overlapping interior, eastern and southern regions.
- Geography therefore provides a useful broad interpretation of the spring clusters, although several spatial outliers confirm that the clustering is determined by fitted extremal dependence rather than spatial proximity alone.
- The dissimilarity matrix for Spring (Supplementary Figure S5) is fairly well block-structured, with intra-cluster dissimilarities generally smaller than inter-cluster dissimilarities.
- As no consistent changepoint is identified for spring, this solution provides a useful stationary seasonal comparison with the temporally divided solutions for winter, summer and autumn.
- The two summer periods retain a common large-scale structure, although a number of stations change membership after 1990.
- During 1960–1990, cluster 1 is concentrated more strongly across northern and eastern Spain, while cluster 2 contains many southern and western stations. Cluster 3 occupies an intermediate and geographically mixed position.
- After 1990, several cluster-2 stations located away from the southern coast move to cluster 1, producing a somewhat clearer geographical division between northern/eastern and southern/western locations.
- The summer reorganisation is less extensive than that observed for winter and appears to be driven more strongly by a limited number of stations.
- As for winter, the post-1990 dissimilarity matrix (Supplementary Figure S3) is less clearly block-structured and has a lower overall dissimilarity scale than the earlier matrix.

- The elbow curves in Figure 7 indicate that $k = 2$ is also a plausible representation of the summer structure. In the two-cluster solution, the members of the intermediate cluster are absorbed into cluster 2, indicating weaker separation between these groups than between either group and cluster 1.
- The signed difference matrix between the two summer periods (Supplementary Figure S6) is predominantly positive, providing evidence of an overall reduction in pairwise dissimilarity after 1990.
- Summer contains the largest individual absolute matrix changes among the three divided seasons, with maximum differences of approximately 0.10. These changes are concentrated in several prominent rows and columns rather than being distributed uniformly.
- Donostia/San Sebastián-Igeldo and Granada Base Aérea display particularly pronounced positive bands, indicating that their fitted dependence structures became more similar to those at many other stations after 1990.
- Granada Base Aérea also moves from cluster 2 to cluster 1. This transition is consistent with a weakening of the earlier distinction between its original cluster and the remaining dependence regimes.
- The concentration of the summer changes in a limited number of rows and columns is consistent with the comparatively strong changepoint evidence under the Infinity norm.
- Collectively, the summer results suggest partial convergence of the fitted dependence structures while retaining a broad north/east versus south/west spatial organisation.
- The 1960–1986 autumn clustering resembles the spring solution in retaining a recognisable geographical structure, with distinct groups across western, central and eastern parts of the study region.
- The geographical separation becomes weaker during 1987–2024, with several northern, northeastern, southwestern and Mediterranean stations changing membership.
- This reorganisation is also reflected in the dissimilarity matrices (Supplementary Figure S4), with the later matrix appearing less distinctly block-structured.
- The autumn signed difference matrix is predominantly positive, indicating that many station pairs have smaller fitted dissimilarities after 1986.
- In contrast to summer, the positive autumn differences are distributed relatively widely throughout the matrix, rather than being dominated by only one or two stations.
- Autumn therefore provides evidence of a broad but generally moderate convergence in fitted extremal dependence structure.
- The three seasons with candidate temporal changes exhibit appreciable reorganisation of their estimated spatial clusters, but the form of this reorganisation differs between seasons.
- Winter exhibits the broadest redistribution of cluster membership and a mixture of increasing and decreasing pairwise dissimilarities.
- Summer retains its principal large-scale spatial structure, but contains several large station-specific changes that weaken the separation between some clusters.
- Autumn exhibits a more widespread reduction in pairwise dissimilarity, although its changepoint evidence is weaker and should be regarded as exploratory.
- Across the divided seasons, the later-period dissimilarity matrices generally exhibit weaker block structure and smaller pairwise dissimilarities. This indicates that the fitted conditional extremal dependence structures became less strongly differentiated across Spain.
- Below two lines should possibly should go in discussion

- These results suggest that the regional differentiation of compound temperature–DPI extremes appears to have weakened in the later periods, with locations displaying more similar fitted dependence structures despite retaining some broad geographical organisation.
- One must also be cautious that the periods used to estimate each clustering solution differ, particularly in the case of Winter 1999–2024, providing fewer observations for fitting the conditional extremes models. The increase in uncertainty inherent in using these shorter periods may effect our clustering.

6 | DISCUSSION

ACKNOWLEDGMENTS

We acknowledge the data providers in the ECA&D project (Klein Tank et al. 2002). Data and metadata available at <https://www.ecad.eu>.

Author contributions

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Financial disclosure

None reported.

Conflict of interest

The authors declare no potential conflict of interests.

SUPPORTING INFORMATION

The following supporting information is available as part of the online article:

How to cite this article: Williams K., B. Hoskins, R. Lee, G. Masato, and T. Woollings (2016), A regime analysis of Atlantic winter jet variability applied to evaluate HadGEM3-GC2, *Q.J.R. Meteorol. Soc.*, 2017;00:1–6.

APPENDIX

A SECTION TITLE OF FIRST APPENDIX

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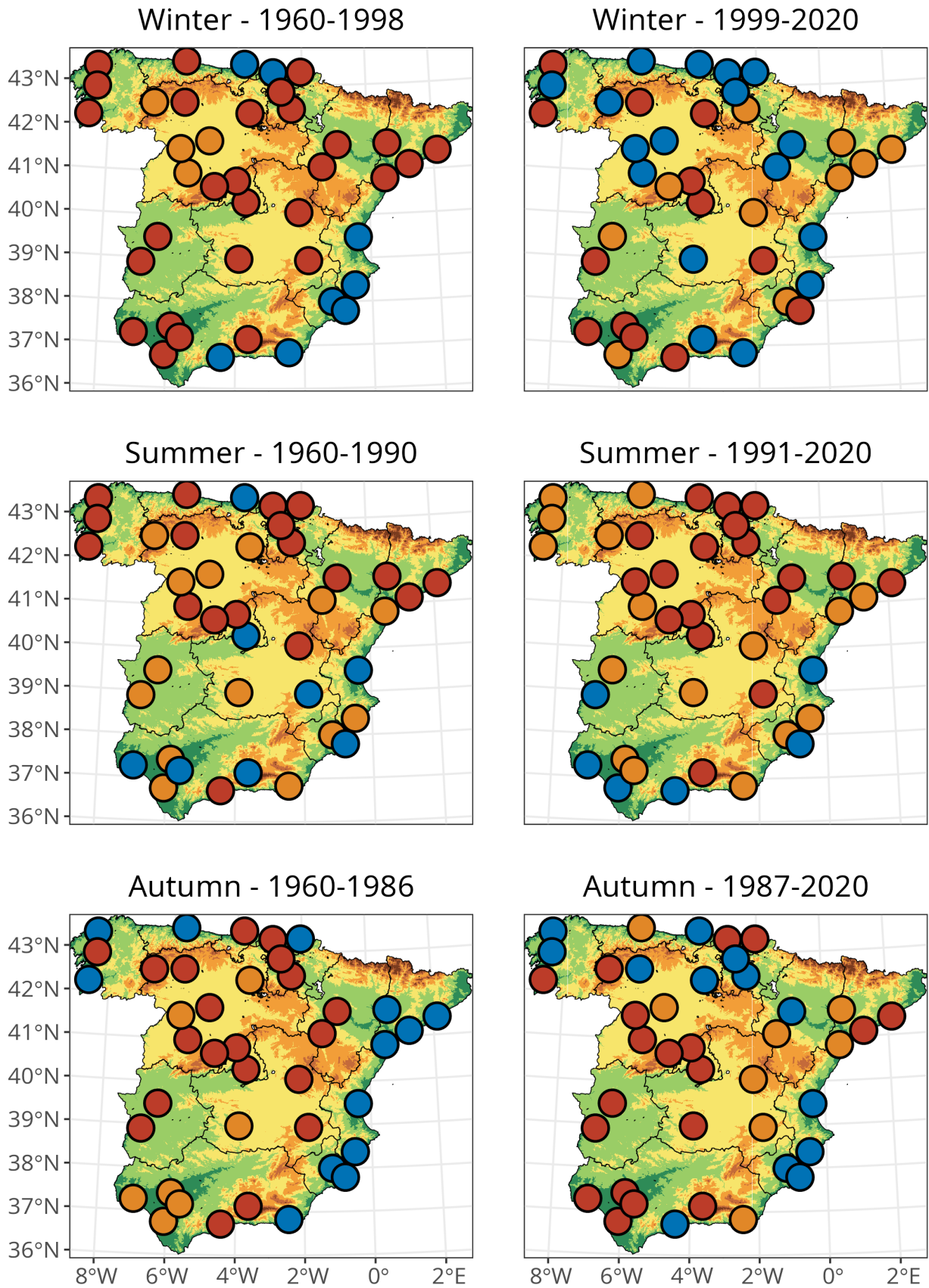


FIGURE 8 Spatial clustering of the 40 Spanish stations obtained from the fitted conditional extremes dissimilarity matrices. Winter is fitted separately for season-years 1960–1998 and 1999–2024, summer is fitted separately for season-years 1960–1990 and 1991–2024, and autumn is fitted separately for season-years 1960–1986 and 1987–2024. Three clusters are used in each fit. Colours distinguish cluster membership within each panel. Background shading represents elevation.